

| Symbol                        | Parameter                                                                         | Conditions                | Min                                         | Typ  | Max | Unit |
|-------------------------------|-----------------------------------------------------------------------------------|---------------------------|---------------------------------------------|------|-----|------|
| I <sub>DDA(SCALER)</sub>      | Scaler static consumption from V <sub>DDA</sub>                                   | BRG_EN=0 (bridge disable) | -                                           | 0.2  | 0.3 | μA   |
|                               |                                                                                   | BRG_EN=1 (bridge enable)  | -                                           | 0.82 | 1   |      |
| t <sub>START_SCALER</sub>     | Scaler startup time                                                               | -                         | -                                           | 140  | 250 | μs   |
| t <sub>START</sub>            | Comparator startup time to reach propagation delay specification                  | High-speed mode           | -                                           | 2    | 5   | μs   |
|                               |                                                                                   | Medium mode               | -                                           | 5    | 20  |      |
| t <sub>D</sub> <sup>(2)</sup> | Propagation delay for 200 mV step with 100 mV overdrive                           | High-speed mode           | -                                           | 50   | 80  | ns   |
|                               |                                                                                   | Medium mode               | -                                           | 0.5  | 0.9 | μs   |
|                               | Propagation delay for step > 200 mV with 100 mV overdrive only on positive inputs | High-speed mode           | -                                           | 50   | 120 | ns   |
|                               |                                                                                   | Medium mode               | -                                           | 0.5  | 1.2 | μs   |
| V <sub>offset</sub>           | Comparator offset error                                                           | Full common mode range    | -                                           | -    | ±20 | mV   |
| V <sub>hys</sub>              | Comparator hysteresis                                                             | No hysteresis             | -                                           | 0    | -   | mV   |
|                               |                                                                                   | Low hysteresis            | -                                           | 10   | -   |      |
|                               |                                                                                   | Medium hysteresis         | -                                           | 20   | -   |      |
|                               |                                                                                   | High hysteresis           | -                                           | 30   | -   |      |
| I <sub>DDA(COMP)</sub>        | Comparator consumption from V <sub>DDA</sub>                                      | Medium mode               | Static                                      | -    | 5   | μA   |
|                               |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -    | 6   |      |
|                               |                                                                                   | High-speed mode           | Static                                      | -    | 70  |      |
|                               |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -    | 75  |      |

1. Refer to [Section 5.3.5: Embedded voltage reference](#).
2. Evaluated by characterization - Not tested in production.

### 5.3.23 Timer characteristics

The parameters given in [Section 5.3.23](#), [Table 66](#), and [Section 5.3.23](#) are specified by design, not tested in production.

Refer to [Table 48. I/O static characteristics](#) for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

**Table 65. TIMx characteristics**

| Symbol                 | Parameter                                    | Conditions                     | Min   | Max                     | Unit <sup>(1)</sup>  |
|------------------------|----------------------------------------------|--------------------------------|-------|-------------------------|----------------------|
| t <sub>res(TIM)</sub>  | Timer resolution time                        | -                              | 1     | -                       | t <sub>TIMxCLK</sub> |
|                        |                                              | f <sub>TIMxCLK</sub> = 144 MHz | 6.9   | -                       | ns                   |
| f <sub>EXT</sub>       | Timer external clock frequency on CH1 to CH4 | -                              | 0     | f <sub>TIMxCLK</sub> /2 | MHz                  |
|                        |                                              | f <sub>TIMxCLK</sub> = 144 MHz | 0     | 72                      |                      |
| Res <sub>TIM</sub>     | Timer resolution                             | TIMx (except TIM2/TIM5)        | -     | 16                      | bit                  |
|                        |                                              | TIM2/TIM5                      | -     | 32                      |                      |
| t <sub>COUNTER</sub>   | 16-bit counter clock period                  | -                              | 1     | 65536                   | t <sub>TIMxCLK</sub> |
|                        |                                              | f <sub>TIMxCLK</sub> = 144 MHz | 0.007 | 455.1                   | μs                   |
| t <sub>MAX_COUNT</sub> | Maximum possible count with 32-bit counter   | -                              | -     | 4, 294, 967, 296        | t <sub>TIMxCLK</sub> |
|                        |                                              | f <sub>TIMxCLK</sub> = 144 MHz | -     | 29.826                  | s                    |

1. TIMx, is used as a general term in which x stands for 1, 2, 5, 6, 7, 8, 12, 15, 16, 17.